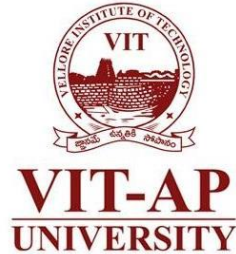


MAT1001 Calculus for Engineers

Line Integrals and Path Independence



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Line Integrals of Scalar Valued Functions - Recap

DEFINITION If C is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \quad (a \leq t \leq b)$$

then

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \|\mathbf{r}'(t)\| dt$$

Similarly, if C is a curve in 3-space that is smoothly parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \quad (a \leq t \leq b)$$

then

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{r}'(t)\| dt$$

DEFINITION If $\mathbf{F}$ is a continuous vector field and C is a smooth oriented curve, then the *line integral of $\mathbf{F}$ along C* is

$$\int_C \mathbf{F} \cdot d\mathbf{r} \quad (28)$$

suppose that C is an oriented curve in the plane given in vector form by

$$\mathbf{r} = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \quad (a \leq t \leq b)$$

If we write

$$\mathbf{F}(\mathbf{r}(t)) = f(x(t), y(t))\mathbf{i} + g(x(t), y(t))\mathbf{j}$$

then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad (29)$$

Formula is also valid for oriented curves in 3-space.

Different ways to write the work integral for $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ over the curve $C: \mathbf{r}(t) = g(t)\mathbf{i} + h(t)\mathbf{j} + k(t)\mathbf{k}$, $a \leq t \leq b$

$$W = \int_C \mathbf{F} \cdot \mathbf{T} \, ds$$

The definition

$$= \int_C \mathbf{F} \cdot d\mathbf{r}$$

Vector differential form

$$= \int_a^b \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} dt$$

Parametric vector evaluation

$$= \int_a^b \left(M \frac{dx}{dt} + N \frac{dy}{dt} + P \frac{dz}{dt} \right) dt$$

Parametric scalar evaluation

$$= \int_C M \, dx + N \, dy + P \, dz$$

Scalar differential form

DEFINITION A vector field $\mathbf{F}$ in 2-space or 3-space is said to be *conservative* in a region if it is the gradient field for some function ϕ in that region, that is, if

$$\mathbf{F} = \nabla\phi$$

The function ϕ is called a *potential function* for $\mathbf{F}$ in the region.

Component Test for Conservative Fields

Let $\mathbf{F} = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$ be a field on a connected and simply connected domain whose component functions have continuous first partial derivatives. Then, $\mathbf{F}$ is conservative if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

A vector field $\mathbf{F}$ defined on a simply connected domain is conservative if $\operatorname{curl} \mathbf{F} = 0$.

Fundamental Theorem of Line Integrals

The following result is the analog of Fundamental Theorem of Calculus for line integrals in 2-space.

THEOREM (*The Fundamental Theorem of Line Integrals*) Suppose that

$$\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$$

is a conservative vector field in some open region D containing the points (x_0, y_0) and (x_1, y_1) and that $f(x, y)$ and $g(x, y)$ are continuous in this region. If

$$\mathbf{F}(x, y) = \nabla\phi(x, y)$$

and if C is any piecewise smooth parametric curve that starts at (x_0, y_0) , ends at (x_1, y_1) , and lies in the region D , then

$$\int_C \mathbf{F}(x, y) \cdot d\mathbf{r} = \phi(x_1, y_1) - \phi(x_0, y_0)$$

or, equivalently,

$$\int_C \nabla\phi \cdot d\mathbf{r} = \phi(x_1, y_1) - \phi(x_0, y_0)$$

Example – The Idea of Path Independence

Stated informally, this theorem shows that *the value of a line integral of a conservative vector field along a piecewise smooth path is independent of the path*; that is, the value of the integral depends on the endpoints and not on the actual path C .

► **Example** Consider the force field $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$

- (a) Confirm that the force field $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$ is conservative by showing that $\mathbf{F}(x, y)$ is the gradient of $\phi(x, y) = xy$.
- (b) Show that the work performed by the field on a particle that moves from the point $(0, 0)$ to the point $(1, 1)$ should be the same along
 - (i) The line segment $y = x$ from $(0, 0)$ to $(1, 1)$.
 - (ii) The parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.
 - (iii) The cubic $y = x^3$ from $(0, 0)$ to $(1, 1)$.
- (c) Use the Fundamental Theorem of Line Integrals to evaluate $\int_{(0,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r}$.

Solution (a). $\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} = y\mathbf{i} + x\mathbf{j}$

Example – The Idea of Path Independence

Solution (b). With $x = t$ as the parameter,

(i) the path of integration is given by $x = t, \quad y = t \quad (0 \leq t \leq 1)$

$$\text{Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (y\mathbf{i} + x\mathbf{j}) \cdot (dx\mathbf{i} + dy\mathbf{j}) = \int_C y dx + x dy = \int_0^1 2t dt = 1$$

(ii) the path of integration is given by $x = t, \quad y = t^2 \quad (0 \leq t \leq 1)$

$$\text{Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C y dx + x dy = \int_0^1 3t^2 dt = 1$$

(iii) the path of integration is given by $x = t, \quad y = t^3 \quad (0 \leq t \leq 1)$

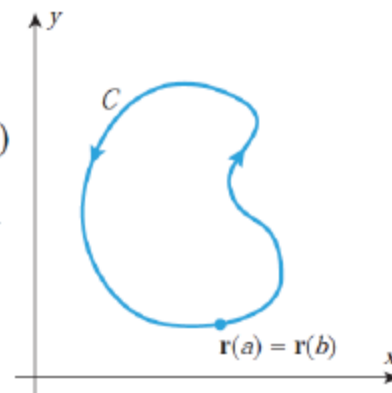
$$\text{Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C y dx + x dy = \int_0^1 4t^3 dt = 1$$

Solution (c).

$$\int_{(0,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{r} = \phi(1, 1) - \phi(0, 0) = 1 - 0 = 1$$

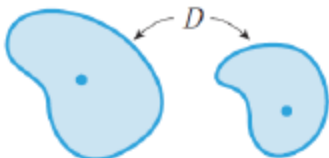
Closed Parametric Curve and Connected Region

A parametric curve C that is represented by the vector-valued function $\mathbf{r}(t)$ for $a \leq t \leq b$ is said to be **closed** if the initial point $\mathbf{r}(a)$ and the terminal point $\mathbf{r}(b)$ coincide; that is, $\mathbf{r}(a) = \mathbf{r}(b)$



Connected

An open region D is said to be **connected**, if any two points in D can be joined by some piecewise smooth curve that lies entirely in D . Stated informally, D is connected if it does not consist of two or more separate pieces



Not connected

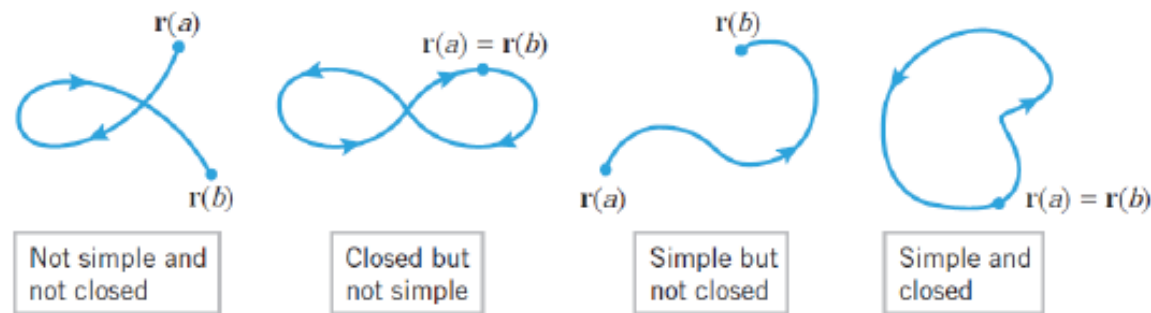
THEOREM *If $f(x, y)$ and $g(x, y)$ are continuous on some open connected region D , then the following statements are equivalent (all true or all false):*

(a) $\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$ is a conservative vector field on the region D .

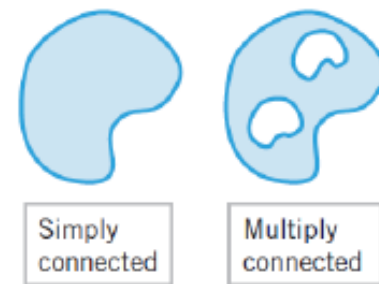
(b) $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every piecewise smooth closed curve C in D .

(c) $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path from any point P in D to any point Q in D for every piecewise smooth curve C in D .

A parametric curve is *simple* if it does not intersect itself between its endpoints. A simple parametric curve may or may not be closed (Figure 15.3.5). In addition, we will say that a connected set D in 2-space is *simply connected* if no simple closed curve in D encloses points that are not in D . Stated informally, a connected set D is simply connected if it has no holes; a connected set with one or more holes is said to be *multiply connected* (Figure 15.3.6).



▲ Figure 15.3.5



▲ Figure 15.3.6

THEOREM (*Conservative Field Test*) If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first partial derivatives on some open region D , and if the vector field $\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$ is conservative on D , then

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x} \quad (9)$$

at each point in D . Conversely, if D is simply connected and (9) holds at each point in D , then $\mathbf{F}(x, y) = f(x, y)\mathbf{i} + g(x, y)\mathbf{j}$ is conservative.

Example

► **Example** Let $\mathbf{F}(x, y) = 2xy^3\mathbf{i} + (1 + 3x^2y^2)\mathbf{j}$.

- (a) Show that $\mathbf{F}$ is a conservative vector field on the entire xy -plane.
- (b) Find ϕ by first integrating $\partial\phi/\partial x$.
- (c) Find ϕ by first integrating $\partial\phi/\partial y$.

Solution (a). Since $f(x, y) = 2xy^3$ and $g(x, y) = 1 + 3x^2y^2$, we have

$$\frac{\partial f}{\partial y} = 6xy^2 = \frac{\partial g}{\partial x}$$

so (9) holds for all (x, y) .

Solution (b). Since the field $\mathbf{F}$ is conservative, there is a potential function ϕ such that

$$\frac{\partial\phi}{\partial x} = 2xy^3 \quad \text{and} \quad \frac{\partial\phi}{\partial y} = 1 + 3x^2y^2 \tag{11}$$

Integrating the first of these equations with respect to x (and treating y as a constant) yields

$$\phi = \int 2xy^3 dx = x^2y^3 + k(y) \quad (12)$$

where $k(y)$ represents the “constant” of integration. We are justified in treating the constant of integration as a function of y , since y is held constant in the integration process. To find $k(y)$ we differentiate (12) with respect to y and use the second equation in (11) to obtain

$$\frac{\partial \phi}{\partial y} = 3x^2y^2 + k'(y) = 1 + 3x^2y^2$$

from which it follows that $k'(y) = 1$. Thus,

$$k(y) = \int k'(y) dy = \int 1 dy = y + K$$

where K is a (numerical) constant of integration. Substituting in (12) we obtain

$$\phi = x^2y^3 + y + K$$

Example

Solution (c). Integrating the second equation in (11) with respect to y (and treating x as a constant) yields

$$\phi = \int (1 + 3x^2y^2) dy = y + x^2y^3 + k(x) \quad (13)$$

where $k(x)$ is the “constant” of integration. Differentiating (13) with respect to x and using the first equation in (11) yields

$$\frac{\partial \phi}{\partial x} = 2xy^3 + k'(x) = 2xy^3$$

from which it follows that $k'(x) = 0$ and consequently that $k(x) = K$, where K is a numerical constant of integration. Substituting this in (13) yields

$$\phi = y + x^2y^3 + K$$

which agrees with the solution in part (b).

Example – Why Simply Connectedness is Important?

EXAMPLE

Show that the vector field

$$\mathbf{F} = \frac{-y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j} + 0\mathbf{k}$$

satisfies the equations in the Component Test, but is not conservative over its natural domain. Explain why this is possible.

Solution We have $M = -y/(x^2 + y^2)$, $N = x/(x^2 + y^2)$, and $P = 0$. If we apply the Component Test, we find

$$\frac{\partial P}{\partial y} = 0 = \frac{\partial N}{\partial z}, \quad \frac{\partial P}{\partial x} = 0 = \frac{\partial M}{\partial z}, \quad \text{and} \quad \frac{\partial M}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial N}{\partial x}.$$

So it may appear that the field $\mathbf{F}$ passes the Component Test.

Example – Why Simply Connectedness is Important?

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}, 0 \leq t \leq 2\pi.$$

$$d\mathbf{r}/dt = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j}$$

$$\mathbf{F} = \frac{-y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j} = \frac{-\sin t}{\sin^2 t + \cos^2 t} \mathbf{i} + \frac{\cos t}{\sin^2 t + \cos^2 t} \mathbf{j} = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j}.$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} dt = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = 2\pi.$$

Since the line integral of $\mathbf{F}$ around the loop C is not zero, the field $\mathbf{F}$ is not conservative

Explain why this is possible

the domain of $\mathbf{F}$ is not simply connected,

Since $x^2 + y^2$ cannot equal zero, the natural domain is the complement of the z -axis

EXAMPLE Show that $\mathbf{F} = (e^x \cos y + yz)\mathbf{i} + (xz - e^x \sin y)\mathbf{j} + (xy + z)\mathbf{k}$ is conservative over its natural domain and find a potential function for it.



Thank You...

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